

深圳南泰微科技有限公司

Shenzhen Nantaiwei Technology Co.,Ltd

SPECIFICATION

规格书

Product Type: 4.0 寸 TFT 液晶显示模块
Model NO: NTW400T51X002A
Customer NO: _____

客户： 客户型号：			产品型号： NTW400T51X002A		
批准 APPROVED	审核 CHECKED	拟制 DESIGNED	批准 APPROVED	审核 CHECKED	拟制 DESIGNED

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变更记录表 Revision				
变更序号 Item NO.	日期 Date	内容及理由 Subject And Reason	版本号 Version	负责人 Response
1	2023.09.25	首版发布 First Issue	V1	TangLIHua

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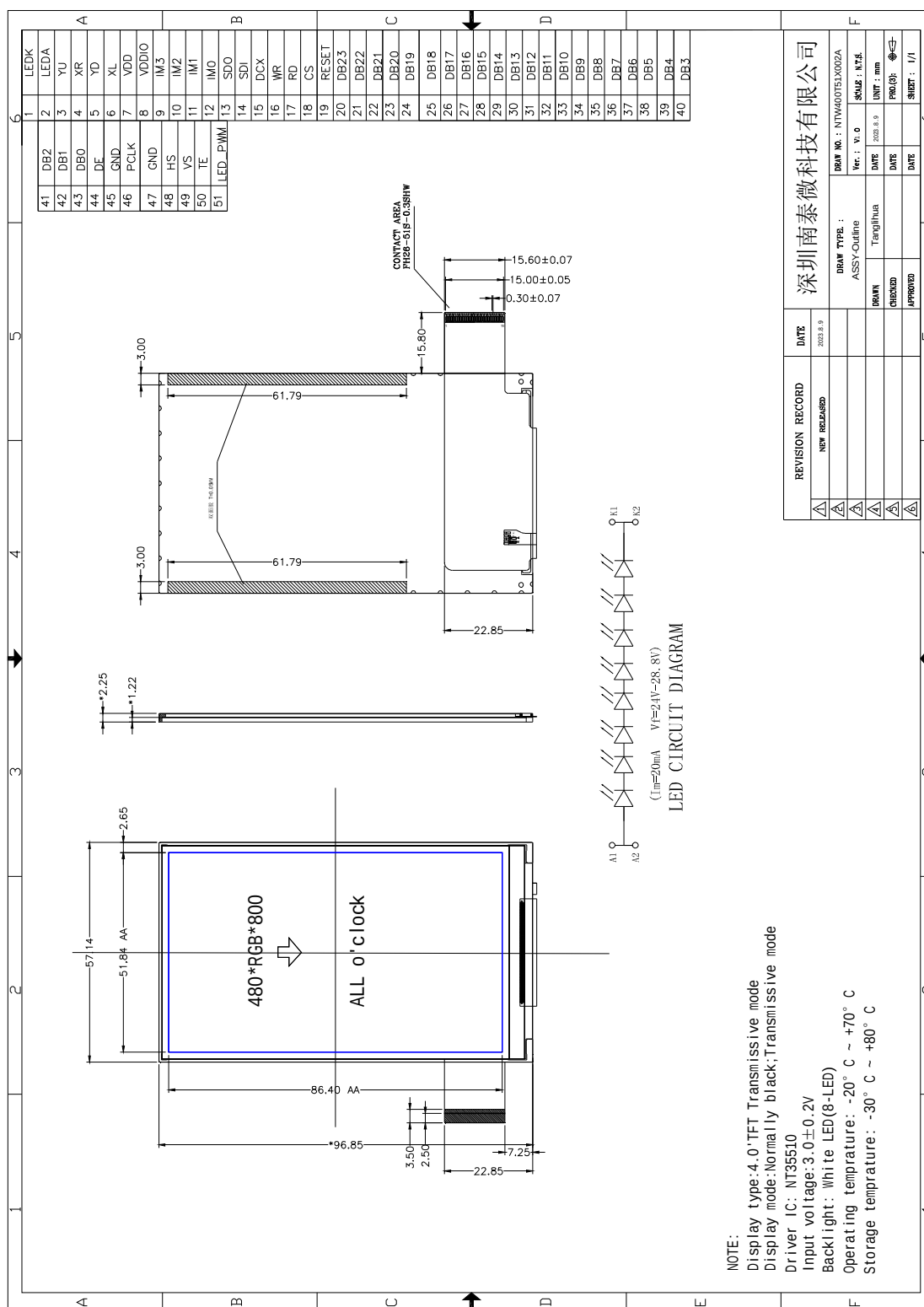
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1. GENERAL INFORMATION 主要特征描述

项目 (Item)	规格 (Specification)	单位 (Units)
LCD Type 液晶显示类型	TFT/TRANSMSSIVE	--
Viewing Direction 视角方向	ALL	0' Clock
Outline Dimension(W*H*T) 外形尺寸 (长宽厚)	57.14*96.85*2.25	mm
Display Area 显示区域	51.84*86.40	mm
Number of Dots 点阵	480*800	--
Driver IC 驱动IC	NT35510H	--
Interface type 接口类型	MCU 8/16bit	--
Input Votage 输入电压	3.3 V	v
Module Power Consumption 模块功耗	—	MW
Colors 色彩	16.7M	--
Backlight type 背光类型	LED	--
模组亮度	400-450	Cd/m ²

Note: COG: Chip On Glass (LCD+IC); BL: Backlight; TP: Touch Panel

2. OUTLINE DRAWING 外形尺寸



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3. INTERFACE DESCRIPTION 接口定义描述

PIN NO	Symbol	I/O	function
1	LEDK		LED Cathode
2	LEDA		LED Anode
3	YU		Touch Panel Control Pin
4	XR		Touch Panel Control Pin
5	YD		Touch Panel Control Pin
6	XL		Touch Panel Control Pin
7	VDD		Power supply (2.8~3.3V)
8	VDDIO		Power supply (1.65~3.3V)
9	IM3		only for SPI,if not used please connect to VSS
10	IM2		IM2=0;IM1=0;IM0=0 8bit MCU
11	IM1		IM2=0;IM1=0;IM0=1 16bit MCU
12	IM0		IM2=0;IM1=1;IM0=1 RGB+SPI
13	SDO		Serial data output pin and used for the DBI type cmode
14	SDI		Serial data input pin and used for the DBI type C mode
15	DCX		Command/Data Select
16	WR		Write signal
17	RD		Read signal
18	CS		Chip Select
19	RESET		LCD RESET TERMINAL ACITVE" L"
20-43	DB23-DB0		DATA BUS
44	DE		RGB I/F Data Enable signal
45	GND		Ground
46	PCLK		RGB I/F Dot Clock Signa
47	GND		Ground
48	HS		RGB I/F Line synchronous signal
49	VS		RGB I/F Frame synchronous signal
50	TE		Output a frame head pulse signal
51	LED_PWM		Output a frame head pulse signal

Note: The voltage power of the interface logic pin depend on IOVCC and GND, Such as DBn, IMn and function pins; Connector : FH26W-51S-0.3SHW

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4. ABSOLUTE MAXIMUM RATINGS 极限参数

项目 (Item)	Symbol 符号	Min 最小值	Max 最大值	Unit 单位
Supply voltage for logic 逻辑电压	VDD	-0.3	+4.2	V
IO Supply Voltage IO口电压	VDDI	-0.3	+4.0	V
Input voltage 输入电平	VIN	-0.3	VDD+0.3	V
Operating temperature 使用温度	TOP	-30	80	°C
Storage temperature 存储温度	TST	-20	70	°C
Storage Humidity 湿度	RH	10	90%(Max60°C)	RH

Note:

***1. At 25±5°C**

***2. Due to the characteristics of LC Material, the Liquid Crystal driving voltage varies with environmental temperature.**

***3. Non-condensation.**

***4. Temp. ≤ 60°C, 90%RH Max.**

Temp. > 60°C, Absolute humidity shall be less than 90%RH.

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5. ELECTRICAL CHARACTERISTICS 模块电气特性

5.1 DC CHARACTERISTICS 直流特性

项目 (Item)	Symbol 符号	Min 最小值	Typ 典型值	Max 最大值	Unit 单位
Supply voltage for logic	VDD	3.0	3.3	4.2	V
Input current	IDD		TBD	TBD	MA
TFT Gate ON Voltage	VGH	--	-	--	V
TFT Gate OFF Voltage	VGL	--	-	--	V
TFT Common Voltage	VCOM	--	-	--	V

Note:

***1. VGH is TFT Gate operating Voltage.**

***2. VGL is TFT Gate operating Voltage.**

The storage structure of this model is C ST (Storage on Common)

***3. Vcom must be adjusted to optimize display quality _Cross talk, Contrast Ratio and etc.**

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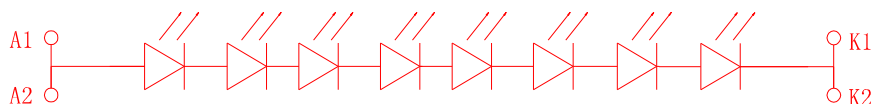
6. BACKLIGHT CHARACTERISTICS 背光电气特性

项目 (Item)	Symbol 符号	Min 最小值	Typ 典型值	Min 最大值	Unit 单位	Condition 条件
Forward voltage 正向电压	VF	-	24	-	V	
LED Current	IF	-	20	-	MA	
Life Time		-	20000	-	Hr	$I \leq 20\text{mA}$

Note (1) Permanent damage may occur to the LCD module if beyond this specification. Functional operation should be restricted to the conditions described under normal operating conditions.

(2) $T_a = 25 \pm 2^\circ\text{C}$

(3) Test condition: LED Current 80mA 供电方式 (Power supply)



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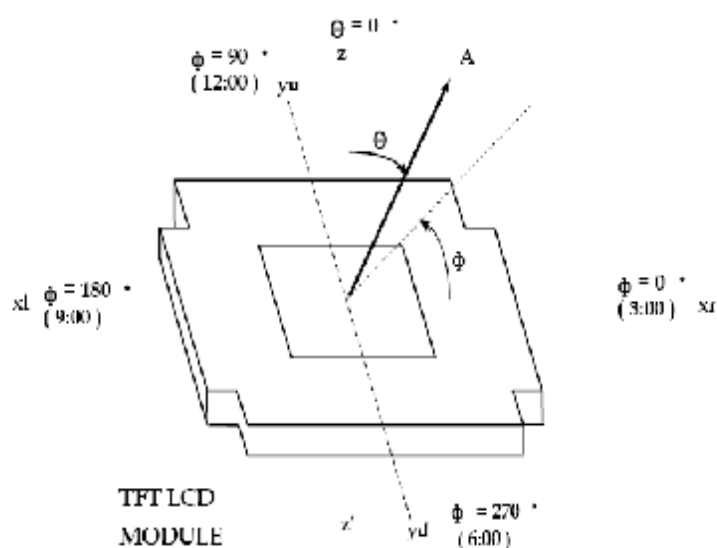
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7. OPTICAL CHARACTERISTICS 光电参数

7.1 Optical Specification

Items		Symbol	Condition	Specifications			Unit	Note
				Min.	Typ.	Max.		
Contrast Ratio(LCM)		C/R		-		-	-	
Response Time		T _R		-	10	20	ms	
		T _F		-	15	20	ms	
Surface Luminance (LCM)		IV		400	450		Cd/m ²	
Chromaticity	Red	X _R		0.604	0.634	0.664	-	
		Y _R		0.298	0.328	0.358	-	
	Green	X _G		0.264	0.294	0.324	-	
		Y _G		0.547	0.577	0.607	-	
	Blue	X _B		0.107	0.137	0.167	-	
		Y _B		0.104	0.134	0.164	-	
	White	X _W		0.272	0.302	0.332	-	
		Y _W		0.305	0.335	0.365	-	
Viewing angle	Hor.	1(3 o'clock)	Center CR≥10	-	80	-	deg.	
		2(9 o'clock)		-	80	-		
	Ver.	2(12 o'clock)		-	80	-		
		1(6 o'clock)		-	80	-		
NTSC ratio					61.5		%	

7.2 Description of View Angle 视角定义



Note: The voltage power of the interface logic pin depend on IOVCC and GND, Such as DBn, IMn and function pins.

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8. REFERENCE APPLICATION CIRCUIT 参考应用电路

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9. Reliability 可靠性试压

测试项目	测试条件	结果
高温存储	+80℃±2℃ 96hr	1. 外观无缺陷 2. 显示无异常
低温存储	-30℃±2℃ 96hr	
高温运行 (HTS)	+60℃±2℃ 24hr	
低温运行 (LTS)	-20℃±2℃ 24hr	
高温高湿运行 (THO)	60℃±2℃ 90%RH 96hr	
冷热循环 (TST)	-30℃ ↔25℃↔ 80℃ 5min 30min ↔25℃ , 5min after 10cycle,	
震动试验 (包装状态下)	10Hz~150Hz, 100m/s2, 120min	

Note: (1) All tests above are practiced at module type.

(2) There is no display function NG issue occurred, all the cosmetic specification is judged before the reliability stress.

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10. INSPECTION CRITERION 检查标准

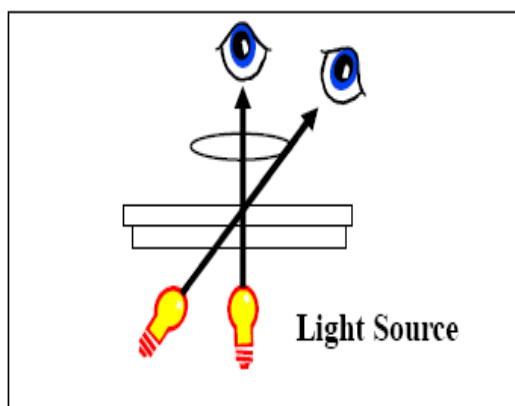
10.1 Sampling Method

Unless otherwise agreed upon in writing, the sampling inspection shall be applied to the Customer's incoming inspection.

- 1) Lot size: Quantity per shipment lot
- 2) Sampling type: Normal inspection , single sampling
- 3) Inspection level: II
- 4) Sampling table: MIL-STD-105D
- 5) Acceptable Quality Level(AQL): Major=0.65 Minor=1.5

10.2 Sampling Method

- 1) Ambient Condition:
 - a. Temperature: Room temperature $25\pm 5^{\circ}\text{C}$
 - b. Illumination: Single fluorescent lamp non-directive(300 to 700 Lux)
- 2) Viewing distance
The distance between the LCD and the inspector's eyes shall be at least 30-50cm.
- 3) Viewing Angle
The inspection shall be conducted within normal viewing angle range.



10.3 Major defect

No.	Item	Inspection Standard	Classification of defects
1	All functional defects	1) No display 2) Display abnormally 3) Open or missing segment 4) Short circuit	Major

13

/17

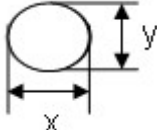
RoHS

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		5) Excess power consumption 6) Backlight no lighting, flickering and abnormal lighting	
2	Missing	Missing component	Major
3	Outline dimension	Overall outline dimension beyond the drawing is not allowed	Major

10.4 Cosmetic Defect

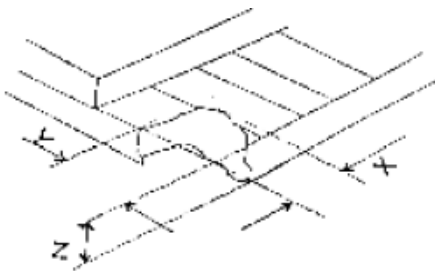
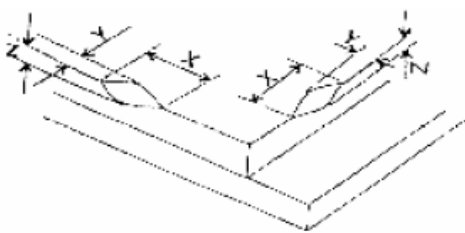
No.	Item	Inspection Standard		Classification of defects										
1	(spot defect) Black and White spot pinhole	For dark/white spot, size Φ is defined as $\Phi=(x+y)/2$		Minor										
2	(line defect) Black line, White line, Foreign Material under Pol&CTP	Define:  <table border="1"> <thead> <tr> <th>Width(mm)</th> <th>Length(mm);Acceptable Qty</th> </tr> </thead> <tbody> <tr> <td>$W\leq0.03$</td> <td>Ignore</td> </tr> <tr> <td>$0.03<W\leq0.05$</td> <td>$L\leq3.0$; $N\leq2$</td> </tr> <tr> <td>$0.05<W\leq0.1$</td> <td>$L\leq2.0$; $N\leq2$</td> </tr> <tr> <td>$0.1<W$</td> <td>Define as spot defect</td> </tr> </tbody> </table>		Width(mm)	Length(mm);Acceptable Qty	$W\leq0.03$	Ignore	$0.03<W\leq0.05$	$L\leq3.0$; $N\leq2$	$0.05<W\leq0.1$	$L\leq2.0$; $N\leq2$	$0.1<W$	Define as spot defect	Minor
Width(mm)	Length(mm);Acceptable Qty													
$W\leq0.03$	Ignore													
$0.03<W\leq0.05$	$L\leq3.0$; $N\leq2$													
$0.05<W\leq0.1$	$L\leq2.0$; $N\leq2$													
$0.1<W$	Define as spot defect													
3	Clear spots Black and white spot defect pinhole, Foreign Particle, Dirt under Pol&CTP.	For dark/white spot, size Φ is defined <table border="1"> <thead> <tr> <th>Size Φ(mm)</th> <th>Acceptable Qty</th> </tr> </thead> <tbody> <tr> <td>$\Phi\leq0.10$</td> <td>Ignor</td> </tr> <tr> <td>$0.10<\Phi\leq0.20$</td> <td>2</td> </tr> <tr> <td>$0.20<\Phi\leq0.30$</td> <td>1</td> </tr> <tr> <td>$0.30<\Phi$</td> <td>0</td> </tr> </tbody> </table>		Size Φ (mm)	Acceptable Qty	$\Phi\leq0.10$	Ignor	$0.10<\Phi\leq0.20$	2	$0.20<\Phi\leq0.30$	1	$0.30<\Phi$	0	Minor
Size Φ (mm)	Acceptable Qty													
$\Phi\leq0.10$	Ignor													
$0.10<\Phi\leq0.20$	2													
$0.20<\Phi\leq0.30$	1													
$0.30<\Phi$	0													

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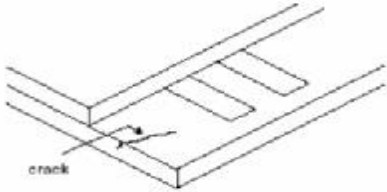
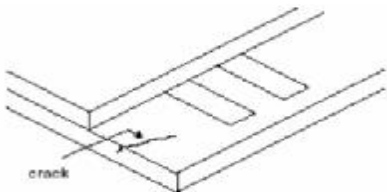
4			
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10.5 Cosmetic Defect

No.	Item	Inspection Standard	Classification of defects						
1	Glass defect	1) Chip on the corner  <table><tr><td>X</td><td>Y</td><td>Z</td></tr><tr><td>≤3.0</td><td>≤S</td><td>≤T</td></tr></table> Remark: S=contact pad length; T=the thickness of glass Chips on the corner of terminal shall not be allowed to extend into the ITO pad or expose perimeter seal. Acceptable Quantity N≤2.	X	Y	Z	≤3.0	≤S	≤T	Minor
		X	Y	Z					
		≤3.0	≤S	≤T					
2) Chip on the edge of glass  <table><tr><td>X</td><td>Y</td><td>Z</td></tr><tr><td>Ignore</td><td>≤0.5</td><td>≤T</td></tr></table> Acceptable Quantity: N≤2	X	Y	Z	Ignore	≤0.5	≤T	Minor		
X	Y	Z							
Ignore	≤0.5	≤T							
3) Creak Creaks tend to break are not allowed.	Minor								

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		3) Creak Creaks tend to break are not allowed. 	Minor

11. PRECAUTIONS FOR USE OF LCD MODULES

10.1 Handling Precautions

10.1.1 The display panel is made of glass. Do not subject it to a mechanical shock by dropping it from a high place, etc.

10.1.2 If the display panel is damaged and the liquid crystal substance inside it leaks out, be sure not to get any in your mouth, if the substance comes into contact with your skin or clothes, promptly wash it off using soap and water.

10.1.3 Do not apply excessive force to the display surface or the adjoining areas since this may cause the color tone to vary.

10.1.4 The polarizer covering the display surface of the LCD module is soft and easily scratched.

Handle this polarizer carefully.

10.1.5 If the display surface is contaminated, breathe on the surface and gently wipe it with a soft dry cloth. If still not completely clear, moisten cloth with one of the following solvents:

— Isopropyl alcohol

— Ethyl alcohol

Solvents other than those mentioned above may damage the polarizer.

Especially, do not use the following:

— Water

— Ketone

— Aromatic solvents

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10.1.6 Do not attempt to disassemble the LCD Module.

10.1.7 If the logic circuit power is off, do not apply the input signals.

10.1.8 To prevent destruction of the elements by static electricity, be careful to maintain an optimum work environment.

a. Be sure to ground the body when handling the LCD Modules.

b. Tools required for assembly, such as soldering irons, must be properly ground.

c. To reduce the amount of static electricity generated, do not conduct assembly and other work under dry conditions.

d. The LCD Module is coated with a film to protect the display surface. Be care when peeling off this protective film since static electricity may be generated.

10.2 Storage precautions

10.2.1 When storing the LCD modules, avoid exposure to direct sunlight or to the light of fluorescent lamps.

10.2.2 The LCD modules should be stored under the storage temperature range. If the LCD modules will be stored for a long time, the recommended condition is:

Temperature : 0℃ ~ 40℃

Relatively humidity: ≤ 80%

10.2.3 The LCD modules should be stored in the room without acid, alkali and harmful gas.

10.3.3 The LCD modules should be no falling and violent shocking during transportation, and also should avoid excessive pressure, water, damp and sunshine.